

SCIENTIFIC ANALYSIS METHODS AND IMPLEMENTATION SPECIFICATION

**Spatial support, temporal change, cross-product observability
and spatio-temporal concentration of landscape fire in Ghana**

Ghana Fire RP1 Analysis - RP1 Analysis v1.1.0
Current final-state scientific methods authority
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Authority item	Current value
Project	Ghana Fire RP1 Analysis
Software release	RP1 Analysis v1.1.0
Distribution	rp1-analysis-v1
Python package	rp1_analysis_v1
Python	>=3.13,<3.14
Analysis schema	rp1-analysis-v1.1
Data schema	rp1-data-schema-v1.1
Canonical notebook	RP1_Analysis_v1.ipynb
Canonical environment	rp_ghana_fire
External cluster engine	SaTScan 10.3.3
Scientific method change	NONE

Document authority. This document describes the current executable product. Scientific intent is expressed here and in the machine-readable contracts; config/analysis_contract.yml is the sole executable owner of study-specific analysis choices. Generic Python implements qualified algorithms and structural validation rather than a competing Ghana-specific study design.

Scientific method change: NONE.

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1. Purpose, scope and authority

The analysis is observational. It characterises remotely observed fire-regime structure and observation-system correspondence across Ghana without estimating causal effects of climate, land use, management, ignition source or other drivers. The product separates MCD64A1 mapped burned-surface evidence from VIIRS thermal-detection evidence and does not treat either satellite product as ground truth for the other.

Authority chain: this methods specification states the current scientific design; config/analysis_contract.yml is its executable study-specific projection; config/data_schema_contract.yml defines data and geometry semantics; config/method_authorities.yml identifies qualified method implementations and parameter provenance; config/output_contract.yml and config/figure_contract.yml define publication outputs and rendering. Scientific run metadata are immutable snapshots and are not rewritten when a current documentation authority is rematerialised without a scientific-method change.

2. Scientific questions

Component	Question	Primary observational target
RQ1	How do magnitude, spatial heterogeneity, seasonal timing and spatial organisation of mapped burned area compare between Ghana agro-climatic-zone and district supports during 2001-2024?	MCD64A1 mapped burned area at direct ACZ and district supports.
RQ2	What are the direction, magnitude and inferential support for monotonic change in annual mapped burned-area rate across five ACZs during 2001-2024?	Five annual ACZ MCD64A1 fixed-support rate series.
RQ3	Among VIIRS-positive district-months, which observed thermal-detection characteristics are associated with absence of corresponding MCD64A1 mapped burned area?	Paired VIIRS/MCD64A1 district-month observations.
Secondary	Where and when do mapped burned-surface events and VIIRS detections form statistically unusual district-level space-time concentrations?	Product-appropriate retrospective SaTScan models.

3. Data authority, product semantics and analytical populations

Runtime analysis uses only staged, hash-qualified consolidated panels and governed geometries shipped with the analysis subproject. Structural missingness and observed zeroes remain distinct. The canonical monthly spine is January 2001 through December 2024; paired VIIRS/MCD64A1 support begins in February 2012.

Analytical population	Rows	Units	Window	Eligibility
ACZ monthly	1,440	5	2001-01 to 2024-12	Complete balanced monthly spine.
District monthly	74,880	260	2001-01 to 2024-12	Complete balanced monthly spine.
Long-run district	72,000	250	2001-01 to 2024-12	Positive fixed BA-2001 burnable-union denominator for all 288 months.
Paired overlap	38,595	249	2012-02 to 2024-12	Positive fixed BA-2012 support and complete paired support for 155 months.
RQ3 model	22,939	249	2012-02 to 2024-12	Paired population restricted to VIIRS-primary detection count > 0.

Construct	Governed field	Role
MCD64A1 BA-2001 area	modis_ba_km2_ba2001	RQ1/RQ2 mapped burned-surface quantity.
Fixed BA-2001 burnable area	modis_burnable_km2_union_ba2001	RQ1/RQ2 denominator; MCD SaTScan exposure.
MCD64A1 BA-2012 presence	modis_ba_any_ba2012	RQ3 outcome component.
VIIRS primary detections	viirs_det_primary	RQ3 focal predictor; VIIRS SaTScan cases.
VIIRS mean FRP	viirs_frp_mean_mw	RQ3 focal predictor.
Fixed BA-2012 burnable area	modis_burnable_km2_union_ba2012	RQ3 log2 adjustment.
VIIRS active days	viirs_days_active_nh	Governed source field; excluded from final RQ3 model.
MODIS active-fire co-detection	modis_det_primary_any	Governed source field; excluded from final RQ3 model and not a validation authority.

4. General analytical and governance principles

- All inferential families, permutation counts, seeds, model terms, factor references and SaTScan settings are specified before result interpretation.
- No parameter is tuned to maximise significance or improve an observed result.
- Alpha is 0.05 and confidence intervals use 95% coverage unless a method-specific rule states otherwise.
- RP1-controlled stochastic procedures use the configured seed 20260822.
- Mathematically or computationally inadmissible analyses fail closed; no silent fallback estimator or ad hoc correction is substituted.
- Publication values are derived from governed analysis authorities rather than manual transcription.
- Interpretation remains observational and does not identify causal effects or drivers.

5. RQ1 - spatial support, heterogeneity, seasonality and spatial organisation

5.1 Spatial supports and rates

RQ1 compares the same MCD64A1 mapped burned-area construct at five directly aggregated ACZ supports and 250 eligible district supports. ACZ values are calculated directly on ACZ support rather than averaged from district rates. Annual rate is $100 \times \text{annual burned area} / \text{fixed BA-2001 burnable area}$. Long-run rate is the arithmetic mean of the 24 annual rates.

5.2 District heterogeneity and departure

Within each parent ACZ, district long-run rates are summarised by median, interquartile range and Gini coefficient. District parent ACZ is assigned by maximum original-polygon overlap share, with whole districts retained. District departure is the raw district long-run rate minus the directly aggregated parent-ACZ long-run rate.

5.3 Circular seasonality

The full 12-month climatology is retained. Circular mean direction describes timing and mean resultant length describes concentration. The reported continuous month coordinate is cyclic at the December/January boundary: 0 is equivalent to 12. This is a circular coordinate, not a linear variable strictly confined to [1,12]. Peak month is descriptive only.

5.4 Spatial organisation

First-order Queen contiguity weights are row-standardised. Islands are excluded from Moran inference without artificial links. Exactly one national Global Moran I is computed with 9,999 permutations. Local Moran uses 9,999 conditional permutations and BH-FDR across eligible non-island districts; only FDR-supported classes are interpreted inferentially.

6. RQ2 - monotonic annual burned-area-rate change under serial dependence

6.1 Question, effect magnitude and inferential target

Each ACZ contributes one annual fixed-support burned-area-rate series for 2001-2024 ($n=24$). Sen slope reports effect direction and magnitude. The inferential procedure is the Romano-Tirlea studentized global Mann-Kendall permutation test. Sen slope and the inferential null are distinct: the inferential null is not defined as a zero Sen slope.

In the supplied reference framework, the global null is strict stationarity. The stated asymptotic validity conditions include strict stationarity, absolute regularity, no ties/continuous marginal distribution, summable beta-mixing coefficients, positive long-run variance and an asymptotically admissible bandwidth sequence. A non-significant result is therefore described as non-rejection under the configured procedure; it does not prove that no temporal change occurred.

6.2 Tie admissibility

The strict production route requires tie-free realised annual observations. All five Ghana ACZ annual series contain 24 distinct values. If a future governed annual series contains ties, production inference fails closed with a method-admissibility error before permutation inference. Generic Mann-Kendall pairwise code retains the mathematically correct zero contribution for an equal pair, but this local kernel behaviour is not inferential qualification for tied time series. The project does not claim that the Romano-Tirlea reference theorem validates a tied-data extension without a separate prospective authority.

6.3 Studentized statistic and finite-sample implementation

The implementation uses the right-continuous less-than-or-equal empirical CDF, a long-run variance estimate with configured bandwidth, and re-studentises every sampled permutation. Two-sided extremeness is based on absolute studentized statistics with equality included in the $\geq$ tail. The independent reference implementation uses exact tie-free rank arithmetic at this boundary.

Component	Current authority	Provenance classification
Global Mann-Kendall rank statistic	Romano-Tirlea global construction	method-derived
Studentisation and long-run variance	Romano-Tirlea construction	method-derived
Per-permutation re-studentisation	Studentized permutation construction	method-derived
Bandwidth rule	$\text{floor}(n^{1/3})$; realised $b=2$ at $n=24$	study-design defined
Variance floor	0.001	study-design defined
Permutation count	9,999	study-design defined
Alternative	two-sided absolute extremeness	study-design defined
Seed	20260822	computational reproducibility
Monte Carlo p-value	plus-one: $(\text{extreme}+1)/(B+1)$	study-design defined

Component	Current authority	Provenance classification
Multiplicity	Benjamini-Hochberg over exactly five ACZ raw p-values	study-design defined

The supplied Romano-Tirlea paper uses $\text{floor}(n^{1/3})$ and $\text{epsilon}=0.001$ in simulations as practical choices; Theorem 3.4 admits a broader class of asymptotically admissible bandwidth sequences and does not uniquely mandate those finite-sample choices. The plus-one Monte Carlo rule is a governed project choice rather than being attributed solely to that theorem.

6.4 Realised Ghana RQ2 result state

ACZ	Sen slope / year	Extreme	Raw p	BH q	BH q < 0.05
Coastal Zone	-0.1647449475	2321	0.2322	0.290250	No
Forest Zone	-0.0531408764	5740	0.5741	0.574100	No
Guinea Savannah	-0.4502995257	832	0.0833	0.208250	No
Sudan Savannah	-0.5015820537	1371	0.1372	0.228667	No
Transition Zone	-0.6753271232	570	0.0571	0.208250	No

All five Sen slopes are negative, but none has BH-FDR-supported evidence of monotonic decline at $\alpha 0.05$. This is an observational inferential statement, not a causal conclusion.

7. RQ3 - observation-system correspondence in VIIRS-positive district-months

7.1 Population and outcome

The descriptive paired population contains 38,595 district-months across 249 districts from February 2012 to December 2024. Primary regression is restricted to 22,939 VIIRS-positive district-months. $Y=1$ denotes VIIRS positive with MCD64A1 mapped burned area absent; $Y=0$ denotes VIIRS positive with mapped burned area present. The estimand concerns observation-system correspondence, not the probability that a true fire was missed.

7.2 Predictors and design

Role	Field/coding
Focal	$\log_2(\text{viirs_det_primary})$
Focal	$\log_2(\text{viirs_frp_mean_mw})$
Adjustment	$\log_2(\text{modis_burnable_km2_union_ba2012})$
Adjustment	Month categorical; January reference
Adjustment	Year categorical; 2013 reference
Adjustment	ACZ categorical; Coastal Zone/CZ reference
Excluded predictors	$\text{viirs_days_active_nh}$; $\text{modis_det_primary_any}$

The realised model matrix has 31 columns: intercept, three continuous terms, 11 month indicators, 12 year indicators and four ACZ indicators. VIF is diagnostic only and has no variable-deletion authority.

7.3 Estimator and inference

The estimator is Firth-type penalised binomial-logit GEE with district clustering and working independence. Covariance is the Morel-Bokossa-Neerchal corrected cluster sandwich. Coefficient inference uses standard-normal Wald statistics and intervals. Standardised probabilities are computed at observed Q25/Q75 values for each focal predictor while retaining observed values of other covariates and averaging fitted probabilities over the model population.

8. Secondary analysis - retrospective SaTScan concentration

Scenario	Model	Population	Cases	Exposure	Window
MCD64A1_POISSON_PRIMARY	Retrospective space-time discrete Poisson	250 districts	modis_burned_pixel_count_ba2001	fixed BA-2001 burnable area	2001-01 to 2024-12
VIIRS_STP_PRIMARY	Retrospective space-time permutation	249 districts	viirs_det_primary	None	2012-02 to 2024-12

Common settings are SaTScan 10.3.3, monthly time, circular spatial windows, high clusters only, 50% maximum spatial size, 12-month maximum temporal length, 999 Monte Carlo replications, alpha 0.05, hierarchical geographically non-overlapping reporting and no Gini-optimised reporting. Coordinates use EPSG:32630 and select polygon centroid when covered, otherwise a point-on-surface. No unsupported user-defined SaTScan random seed is passed.

8.1 External execution boundary

Section 04-01 always generates deterministic case, coordinate, model-appropriate exposure and parameter files. The user executes SaTScan externally. Section 04-02 detects, validates and ingests existing result families only. The core result family is the exact extensionless ResultsFile plus ResultsFile.col.txt and ResultsFile.gis.txt; no duplicate ResultsFile.txt and no external provenance JSON are mandatory. With neither result family present the state is EXTERNAL_RESULTS_REQUIRED. Two complete, current-run, provenance-bound and parse-valid families produce RESULTS_VALIDATED. Partial, stale, malformed, path-escaping or identity-mismatched material fails closed. Section 90 always executes according to this state.

9. Software implementation and execution workflow

RP1_Analysis_v1.ipynb is a thin orchestrator. Package-owned code implements statistics, spatial analysis, PGEE/MBN estimation, SaTScan interface generation/validation, publication construction and provenance. The notebook loads configuration and governed inputs, invokes package builders and records run metadata.

1. Load and validate the five executable configuration contracts.
2. Validate governed panels, geometries, keys, support and relationships.
3. Build RQ1 authorities and presentation sources.
4. Build five annual RQ2 series and execute the tie-admissible studentized permutation inference.
5. Build RQ3 paired/model populations, diagnostics, 31-column design, PGEE/MBN fit and standardised probabilities.
6. Generate deterministic SaTScan 04-01 interfaces and external-execution manifests.
7. Validate optional 04-02 result state without launching SaTScan.
8. Construct all publication products allowed by the current state and validate run manifests/hashes.

10. Publication outputs

ID	Current content / availability
T1	ACZ RQ1/RQ2 summary including direct ACZ rate, district heterogeneity/circular summaries, Sen slope, raw RQ2 p and BH q.
T2	RQ3 continuous model summary: VIIRS count, mean FRP and burnable-area adjustment.
T3	Significant validated SaTScan clusters; requires genuine RESULTS_VALIDATED.
F2	Spatial supports and long-run mapped burned-area burden.
F3	RQ1 seasonality and spatial organisation.
F4	Five ACZ annual fixed-support trajectories with Sen slope, raw p and BH q.
F5	RQ3 cross-product observability.
F6	Descriptive recurrence: number of distinct validated significant clusters containing each eligible district; requires RESULTS_VALIDATED.

ID	Current content / availability
S1-S6	Supplementary tables for populations, district RQ1, spatial inference, RQ2 annual/inference authority, RQ3 diagnostics/full coefficients and SaTScan results.
Figure S1	Full district seasonal diagnostics.

11. Reproducibility and qualification requirements

The canonical environment is `rp_ghana_fire` with Python $\geq 3.13, < 3.14$. Source qualification verifies that `rp1_analysis_v1`, `mv_firms_panels` and `geo_data_prep` import from the current repository before running the complete test estates. Governed inputs and active reference methods must pass validation before analysis acceptance.

```
python tools/qualify_source_checkout.py
python RP1_Project/rp1_analysis/scripts/validate_inputs.py
python RP1_Project/rp1_analysis/scripts/qualify_reference_methods.py
python RP1_Project/rp1_analysis/scripts/execute_notebook.py --run-id <run_id>
python RP1_Project/rp1_analysis/scripts/validate_run.py --run-id <run_id> --write-report
```

Runs record configuration hashes, governed input identities, realised sample sizes/parameters, method qualification, output inventories and SHA-256 hashes. Current human-readable methods documents are repository-owned under `docs/` and their exact bytes are bound by `data/authorities/methods_document_authority.json`. Historical scientific run metadata are not rewritten by a documentation-only authority update.

12. Interpretation boundaries

- MCD64A1 and VIIRS observe different manifestations of fire and are not interchangeable measurements or ground truth.
- Satellite omission/commission, overpass timing, atmospheric obscuration, canopy/surface conditions, viewing geometry, product resolution and monthly aggregation can contribute to disagreement.
- Districts are reporting units, not causal fire-process units.
- RQ2 has only 24 annual observations per ACZ and inference is tied to the prespecified studentized permutation procedure and its reference scope.
- RQ2 non-rejection does not demonstrate absence of temporal change; Sen slopes are effect estimates and are not causal.
- RQ3 is conditional on VIIRS positivity and estimates marginal associations with mapped-burn absence, not product accuracy or causal mechanisms.
- SaTScan clusters are statistically unusual concentrations under the governed model, not ignition causes, permanent hotspot classifications or causal risk zones.

13. Reference-method scope and parameter provenance

Independent reference fixtures validate numerical implementations without owning the Ghana study design and do not import the production function under test as the reference calculation. For the Romano-Tirlea method, the supplied reference paper explicitly states the global stationarity null, the no-ties condition and weak-dependence/positive-variance/bandwidth conditions. The product therefore distinguishes generic rank-kernel tie behaviour from inferential qualification and fails closed on tied annual RQ2 input.

Method provenance is explicit: method-derived mathematical structure is separated from prospectively governed finite-sample and reproducibility choices. This prevents documentation from overstating what the external theorem uniquely prescribes.

14. References

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- Anselin (1995); Castro and Singer (2006). doi:10.1111/j.1538-4632.1995.tb00338.x.
- Sen (1968), *Journal of the American Statistical Association*. doi:10.1080/01621459.1968.10480934.
- Romano JP, Tirlea MA (2024). Permutation Testing for Monotone Trend. arXiv:2404.06239v2 (10 April 2024). Journal authority recorded in `method_authorities.yml`: doi:10.1111/jtsa.12850.

Benjamini and Hochberg (1995), JRSS B. doi:10.1111/j.2517-6161.1995.tb02031.x.

Mondol and Rahman (2019), Statistics in Medicine. doi:10.1002/sim.8126.

Liang and Zeger (1986), Biometrika. doi:10.1093/biomet/73.1.13.

Morel, Bokossa and Neerchal (2003), Biometrical Journal. doi:10.1002/bimj.200390021.

Kulldorff et al. (2005), PLoS Medicine. doi:10.1371/journal.pmed.0020059.

Kulldorff retrospective space-time discrete Poisson scan statistic.

Appendix A. Fixed parameter register

Area	Parameter	Specification
General	alpha	0.05
General	RP1 stochastic seed	20260822
RQ1	BA denominator	Fixed BA-2001 burnable union
RQ1	Spatial weights	First-order Queen; row-standardised; islands excluded
RQ1	Global Moran	One national result; 9,999 permutations
RQ1	Local Moran	9,999 conditional permutations; BH-FDR
RQ1	Circular month	Cyclic continuous coordinate; 0 equivalent to 12
RQ2	Effect	Sen slope
RQ2	Reference admissibility	Tie-free realised observations; fail closed on ties
RQ2	Bandwidth	$\text{floor}(n^{1/3})$; realised 2 at $n=24$
RQ2	Variance floor	0.001
RQ2	Permutations	9,999; two-sided absolute; plus-one p
RQ2	Multiplicity	BH over exactly five ACZ raw p-values
RQ3	Model	Firth-type penalised binomial-logit GEE
RQ3	Working correlation	Independence
RQ3	Covariance	Morel-Bokossa-Neerchal
RQ3	Inference	Standard-normal Wald
RQ3	Design	31 columns
Secondary	MCD	Retrospective discrete Poisson with fixed BA-2001 exposure
Secondary	VIIRS	Retrospective space-time permutation; no exposure
Secondary	Spatial/temporal	50% spatial; 12 months temporal
Secondary	Monte Carlo	999
Secondary	Reporting	Hierarchical geographically non-overlapping; Gini disabled

Appendix B. Executable authority hashes

Authority	SHA-256
analysis_contract.yml	9692b53595028e3832c0106e3476edf5d83b7d7763830777bf9c97c9629569ec
data_schema_contract.yml	d7b8ed574438804457e50d04b0135066161061976ca4aa8c6f0944b61a163abf

Authority	SHA-256
method_authorities.yml	248bb4aa5f0d05c75fcfd402eb68446184f7486a1060980849c8bf62ab0bdd70
output_contract.yml	ac4559a71526eb37969fdcd1f2572f7868ab8837a1aebc5cad9418eafab74c3d
figure_contract.yml	f6c549f558ac2ce62e27860dfb66ae2235c3e2ffe39faf41e3796eb1f89fa6a9
RP1_Analysis_v1.ipynb	68ea7cf531820161c44967f26c7d220e6fdbbe58800d3f28bf4d139c58c95787